

Exome sequencing and targeted *DHDDS* analysis reveal the molecular architecture of ten unrelated families with autosomal-recessive inherited retinal dystrophies

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Abstract

Purpose. To define the molecular cause of autosomal-recessive inherited retinal dystrophy (arIRD) in ten unrelated probands, including two Ashkenazi-Jewish (AJ) families investigated by targeted *DHDDS* sequencing, and to correlate genotype with full-field electroretinographic (ERG), optical-coherence-tomography (OCT), and Goldmann visual-field (GVF) phenotype.

Methods. Eight probands (F1–F8) underwent whole-exome sequencing (WES; mean on-target coverage 87.4×, 96.1% of bases at ≥ 20× depth, transition/transversion ratio 2.41). Two AJ probands (F9, F10) were screened by Sanger sequencing of all seven coding exons of *DHDDS* (NM_024887.4). Variants were filtered against gnomAD v2.1.1 (allele frequency cut-off 5×10^{-4} for AR genes), and pathogenicity assessed with CADD v1.6 and REVEL v1.3. Phenotyping comprised ISCEV-standard full-field ERG (scotopic 0.01 cd·s·m⁻², scotopic standard flash 3.0 cd·s·m⁻², photopic 3.0 cd·s·m⁻², 30-Hz flicker), spectral-domain OCT (Heidelberg Spectralis, 9-mm radial scan, 1024 A-scans), and GVF with V4e and III4e isopters.

Results. A definitive bi-allelic causal genotype was identified in 8/10 families. Families F7 and F8 each carry two independent candidate loci that cannot be discriminated on present data; both remain **genetically ambiguous**. F9 and F10 both segregate the AJ founder allele *DHDDS* c.124A>G (p.Lys42Glu) in homozygosity. Residual rod b-wave amplitudes on scotopic standard flash ranged from 11.4 μV (F4, *RPE65*) to 78.9 μV (F2, *ABCA4*); the F6 (*CERKL*) family showed an intermediate, well-clustered phenotype.

Conclusion. Combined WES with targeted founder-allele screening yields a diagnosis in 80% of unrelated arIRD probands; F7 and F8 highlight the residual interpretive headroom of WES even after segregation analysis.

1. Introduction

Inherited retinal dystrophies (IRD) collectively affect an estimated 1 in 3,000 individuals worldwide and constitute the leading cause of untreatable blindness in working-age adults. More than 280 genes have been associated with IRD phenotypes; the autosomal-recessive (AR) subset is the most heterogeneous, with > 110 implicated loci. Whole-exome sequencing (WES) achieves a molecular diagnosis in 55–76% of AR probands across published cohorts (range 55.2% [Chen 2013] to 76.4% [Ramachandran 2019]), leaving a substantial residual fraction in which causality cannot be confidently assigned even after dense segregation analysis. In Ashkenazi-Jewish (AJ) populations, the founder allele *DHDDS* c.124A>G (p.Lys42Glu) — first described by Züchner et al. in 2011 — accounts for roughly 1 in 320 AJ retinitis-pigmentosa cases at a population carrier frequency of 0.84% (gnomAD AJ-non-Finnish, 36/4,288 alleles). Targeted Sanger sequencing of *DHDDS* in AJ probands therefore remains a cost-effective complement to WES.

Here we describe ten unrelated AR-IRD pedigrees: eight investigated by WES (F1–F8) and two AJ families (F9, F10) by targeted *DHDDS* sequencing. We integrate dense quantitative phenotyping — ISCEV-standard ERG (four conditions, two eyes, two affected sibs per family in 7/10 pedigrees), OCT central retinal thickness (CRT) and ellipsoid-zone (EZ) width, and GVF isopter areas — to anchor the molecular findings.

2. Materials and Methods

2.1 Subjects. Ten unrelated probands and 41 additional family members (mean pedigree size 5.1, range 3–8) were enrolled between January 2013 and August 2016 under IRB protocol NPM-OPH-2012-118. Mean age at examination of affected individuals was 34.6 ± 14.2 years (range 7–62). Ancestry self-report: 4 European-non-AJ, 2 Ashkenazi-Jewish, 2 South-Asian, 1 East-Asian, 1 West-African.

2.2 WES. Genomic DNA (1.0 μg) was sheared to 180–220 bp on a Covaris LE220 instrument, captured with Agilent SureSelect Human All Exon V6 (60 Mb), and sequenced on an Illumina NovaSeq 6000 (2 × 150 bp). Reads were aligned to GRCh37/hg19 with BWA-MEM v0.7.17, duplicates removed with Picard v2.20.4, base-quality recalibration and variant

calling performed with GATK v4.1.4.1 HaplotypeCaller. Across F1–F8, mean on-target coverage was $87.4 \times$ (range $71.2 \times$ to $108.6 \times$); percentage of bases at $\geq 20\times$ depth averaged 96.1% (range 93.4–98.2%); transition/transversion ratio 2.41 (range 2.36–2.49). After joint genotyping, $31,842 \pm 1,107$ high-quality variants were retained per proband.

2.3 Variant filtering. Variants were filtered against gnomAD v2.1.1 (allele frequency $< 5 \times 10^{-4}$ for recessive models), restricted to known IRD genes (RetNet, retrieved 2016-06-30, 264 genes), and prioritised by CADD-PHRED ≥ 20 and REVEL ≥ 0.50 for missense calls. Candidate variants underwent orthogonal Sanger confirmation and full segregation in all available family members (mean 4.1 individuals/family).

2.4 Targeted DHDDS sequencing (F9, F10). All seven coding exons plus ≥ 30 bp of intronic flanking sequence were amplified in seven amplicons (sizes 312, 408, 487, 521, 366, 298, 441 bp; total target 2,833 bp) using primer pairs listed in Supplementary Table S1, and sequenced bidirectionally on an ABI 3730 with BigDye Terminator v3.1. Mean Phred quality across the 2,833 bp target was 47.8 (range 39.1–58.6).

2.5 Phenotyping. Full-field ERG followed ISCEV 2015 standards (Diagnosys Espion E3, Ganzfeld stimulator, 30-min dark adaptation). Four conditions were recorded: (i) scotopic dim flash $0.01 \text{ cd}\cdot\text{s}\cdot\text{m}^{-2}$; (ii) scotopic standard flash $3.0 \text{ cd}\cdot\text{s}\cdot\text{m}^{-2}$; (iii) photopic single-flash $3.0 \text{ cd}\cdot\text{s}\cdot\text{m}^{-2}$ on $30 \text{ cd}\cdot\text{m}^{-2}$ background; (iv) 30-Hz flicker. A- and b-wave amplitudes (μV) and implicit times (ms) were measured. OCT acquired 9-mm radial scans (Heidelberg Spectralis, 1024 A-scans, ART 25); central retinal thickness (CRT) was measured at the foveal centre and ellipsoid-zone (EZ) preservation width manually delineated by two graders (intraclass correlation 0.94). GVF used a Haag-Streit 930 perimeter with V4e and III4e isopters; areas were planimetered in square degrees.

2.6 Statistics. Continuous variables are summarised as mean $\pm$ SD; ERG amplitudes are reported as the per-family mean across affected individuals and across the two eyes (i.e. 2–4 measurements per family per condition). All p-values are two-sided; $\alpha = 0.05$. Analyses were performed in R v3.6.3.

3. Results

Table 1. Demographic and basic clinical features of the ten unrelated probands and their affected sibs (n = 21 affected individuals in total across the cohort). Age refers to age at examination, in years. Onset = patient-reported age at first symptom (nyctalopia or central vision loss). LogMAR = best-corrected visual acuity of the proband (better eye). Refraction is spherical equivalent of the proband.

Family	Ancestry	Sex (P)	# Aff.	Age P (y)	Onset (y)	LogMAR	Refraction (D)	Consang.
F1	European	M	2	41	9	0.48	-3.25	No
F2	European	F	3	27	14	0.30	-1.00	No
F3	South-Asian	M	2	16	4	0.78	+0.50	Yes (1°)
F4	European	F	1	7	2	1.30	+4.75	No
F5	West-African	M	2	33	11	0.60	-5.50	No
F6	South-Asian	F	3	29	8	0.54	-2.25	Yes (2°)
F7	European	M	2	45	18	0.40	-0.75	No
F8	East-Asian	F	2	38	12	0.52	-2.00	No
F9	Ashkenazi Jewish	M	2	52	22	0.70	-3.00	No
F10	Ashkenazi Jewish	F	2	48	20	0.66	-2.50	No
Mean / total	—	—	21	33.6	12.0	0.628	-1.50	—

Table 2. Whole-exome sequencing quality-control metrics for probands F1–F8. On-target = within Agilent SureSelect V6 bait intervals $\pm$ 100 bp. Ti/Tv = transition/transversion ratio after VQSR. Het/Hom = ratio of heterozygous to homozygous non-reference variants. F9 and F10 were not subjected to WES; their *DHDDS*-targeted metrics are reported in Table 7.

Proband	Reads ($\times 10^6$)	% mapped	Mean cov.	% $\geq 20\times$	% $\geq 30\times$	Ti/Tv	Het/Hom	# HQ variants	# IRD-gene
F1	112.4	99.31	82.6 $\times$	95.8	91.2	2.41	1.62	31,418	138
F2	138.7	99.42	108.6 $\times$	98.2	96.1	2.49	1.58	32,807	152
F3	98.6	99.17	71.2 $\times$	93.4	87.9	2.38	1.66	30,994	129
F4	121.5	99.28	91.4 $\times$	96.6	92.8	2.43	1.61	31,886	141
F5	109.2	99.05	79.8 $\times$	94.7	89.4	2.36	1.70	30,621	126
F6	127.8	99.36	94.7 $\times$	97.1	93.5	2.45	1.59	32,140	147
F7	115.6	99.21	85.3 $\times$	96.2	91.7	2.42	1.63	31,755	144
F8	131.2	99.39	99.1 $\times$	97.8	94.8	2.47	1.57	32,463	149
Mean	119.4	99.27	87.4 $\times$	96.1	92.2	2.41	1.62	31,761	140.8

Table 3. Causal and candidate variants identified in the ten pedigrees. AF_{gnomAD} = global allele frequency in gnomAD v2.1.1 (NA when absent). CADD = PHRED-scaled CADD v1.6 score. REVEL is reported only for missense variants. # seg. = number of informative meioses in which the variant co-segregated with disease status. For families F7 and F8, two independent candidate loci passed all filtering criteria and remain genetically ambiguous on present data (see S3.4).

Family	Gene	Transcript	cDNA	Protein	Zyg.	AF gnomAD	CADD	REVEL	# seg.	ACMG
F1	USH2A	NM_206933.4	c.2299delG	p.Glu767Serfs*21	hom	1.7e-4	35	—	6/6	P
F2	ABCA4	NM_000350.3	c.5882G>A	p.Gly1961Glu	hom	3.4e-4	27	0.71	7/7	P
F3	CRB1	NM_201253.3	c.2843G>A	p.Cys948Tyr	comp het	2.1e-4	32	0.84	4/4	P
F3	CRB1	NM_201253.3	c.498_506del	p.Ile167_Gly169del	comp het	NA	26	—	4/4	LP
F4	RPE65	NM_000329.3	c.1102T>C	p.Tyr368His	hom	8.0e-5	29	0.78	3/3	P
F5	EYS	NM_001142800.2	c.4350_4356del	p.Ile1451Profs*3	hom	NA	34	—	5/5	P
F6	CERKL	NM_201548.5	c.769C>T	p.Arg257*	hom	4.2e-5	37	—	6/6	P
F7 (cand. A)	PDE6A	NM_000440.3	c.998+1G>A	splice donor	hom	NA	33	—	3/3	P (cand.)
F7 (cand. B)	NR2E3	NM_014249.4	c.166G>A	p.Gly56Arg	hom	6.6e-4	26	0.62	3/3	LP (cand.)
F8 (cand. A)	TULP1	NM_003322.6	c.1495+1G>A	splice donor	hom	NA	31	—	3/3	P (cand.)
F8 (cand. B)	FAM161A	NM_001201543.2	c.1567C>T	p.Arg523*	hom	1.1e-4	36	—	3/3	LP (cand.)
F9	DHDDS	NM_024887.4	c.124A>G	p.Lys42Glu	hom	9.6e-4 (AJ 8.3e-3)	24	0.66	4/4	P
F10	DHDDS	NM_024887.4	c.124A>G	p.Lys42Glu	hom	9.6e-4 (AJ 8.3e-3)	24	0.66	4/4	P

3.1 Molecular findings in F1–F6. WES identified a single convincing bi-allelic causal genotype in each of families F1–F6 (Table 3). The *USH2A* frameshift c.2299delG (F1, ACMG class P; 4 published homozygous probands in our internal database, 27 in LOVD as of 2016-06) explained the Usher-syndrome-compatible phenotype. The *ABCA4* p.Gly1961Glu allele in F2, although often associated with mild phenotypes, segregated bi-allelically in three affected sibs (ages 27, 24 and 19 years). The *CRB1* compound-heterozygous genotype in F3 was confirmed by parental phase: the father transmits c.2843G>A; the mother transmits the in-frame deletion c.498_506del. The *RPE65* p.Tyr368His homozygote in F4 (proband age 7) showed early-onset disease compatible with Leber congenital amaurosis type 2. The *EYS* frameshift in F5 was identified in two affected brothers (ages 33 and 31). The *CERKL* nonsense p.Arg257* in F6 was homozygous in three affected female sibs (ages 29, 26 and 24), consistent with the previously reported Spanish/Mediterranean phenotype (Tuson 2004) but, notably, in a South-Asian family.

3.2 Genetic ambiguity in F7. Two candidate variants in F7 passed all filtering criteria. The *PDE6A* splice-donor variant c.998+1G>A is absent from gnomAD, has CADD 33, and co-segregates with disease in all three informative meioses (both affected sibs homozygous, both unaffected parents heterozygous, one unaffected sib heterozygous). However, the *NR2E3* missense p.Gly56Arg (gnomAD AF 6.6×10^{-4} , CADD 26, REVEL 0.62) shows identical segregation. The two genes map to chromosomes 5q32 (*PDE6A*) and 15q23 (*NR2E3*), so the variants are unlinked. Phenotype alone does not discriminate (proband ERG shows generalised rod-cone dysfunction, compatible with either gene). We therefore report F7 as **genetically ambiguous** pending functional studies.

3.3 Genetic ambiguity in F8. Similarly, in F8 the *TULP1* splice-donor variant c.1495+1G>A and the *FAM161A* nonsense p.Arg523* both passed filtering and both segregate (3/3 informative meioses). *TULP1* (6p21.31) and *FAM161A* (2p15) are again unlinked. F8 is reported as **genetically ambiguous**.

3.4 Targeted *DHDDS* findings in F9 and F10. Sanger sequencing of all seven coding exons of *DHDDS* in F9 and F10 revealed the AJ founder allele c.124A>G (p.Lys42Glu) in homozygosity in both probands and in all four affected sibs (two per family). Mean Sanger quality (Phred) across the 2,833-bp target was 47.8. No other coding variant was identified.

Table 4. Mean full-field ERG a- and b-wave amplitudes (μV) of affected individuals in each pedigree, recorded under four ISCEV-standard conditions. Each family's value is the mean across both eyes of all affected individuals (2–4 measurements per family per condition). Conditions: (i) Scotopic dim flash $0.01 \text{ cd}\cdot\text{s}\cdot\text{m}^{-2}$ (rod-isolated, b-wave only); (ii) Scotopic standard flash $3.0 \text{ cd}\cdot\text{s}\cdot\text{m}^{-2}$ (mixed rod–cone response, a- and b-wave); (iii) Photopic single-flash $3.0 \text{ cd}\cdot\text{s}\cdot\text{m}^{-2}$ on $30 \text{ cd}\cdot\text{m}^{-2}$ background (cone-isolated, a- and b-wave); (iv) 30-Hz flicker, peak-to-peak. Healthy adult reference ranges (mean ± 2 SD) from our laboratory normative database ($n = 124$): scotopic dim b-wave $248 \pm 78 \mu\text{V}$; scotopic SF a-wave $232 \pm 64 \mu\text{V}$, b-wave $418 \pm 96 \mu\text{V}$; photopic a-wave $26 \pm 8 \mu\text{V}$, b-wave $116 \pm 32 \mu\text{V}$; 30-Hz flicker $102 \pm 28 \mu\text{V}$. Bold values denote the per-family means used in subsequent statistical comparisons.

Family	Gene (causal)	Scot 0.01 b-wave (μV)	Scot SF a-wave (μV)	Scot SF b-wave (μV)	Photopic a-wave (μV)	Photopic b-wave (μV)	30-Hz flicker (μV)
F1	USH2A	18.6	11.4	29.8	4.7	21.6	16.2
F2	ABCA4	62.1	41.7	78.9	12.4	61.3	47.1
F3	CRB1	9.3	6.8	16.2	3.1	14.7	11.9
F4	RPE65	5.4	4.2	11.4	1.8	9.6	7.3
F5	EYS	21.7	14.9	37.4	5.6	26.2	19.4
F6	CERKL	24.3	17.8	42.7	6.1	29.5	22.8
F7	PDE6A or NR2E3	15.2	9.6	24.1	5.3	23.4	18.6
F8	TULP1 or FAM161A	12.8	8.4	19.7	4.4	19.1	15.0
F9	DHDDS	11.7	7.9	21.3	4.9	20.8	16.4
F10	DHDDS	13.4	8.6	23.6	5.1	22.0	17.1
Cohort mean	—	19.45	13.13	30.51	5.34	24.82	19.18
Cohort SD	—	16.42	10.81	20.23	2.97	14.81	10.78
Normal (mean ± 2 SD)		248 $\pm$ 78	232 $\pm$ 64	418 $\pm$ 96	26 $\pm$ 8	116 $\pm$ 32	102 $\pm$ 28

Table 4a (supplementary). Per-individual ERG amplitudes (μV) for the three affected sibs of family F6, by eye. OD = right eye, OS = left eye. Means are computed across both eyes and then averaged across sibs to yield the family-level value reported in Table 4. For the scotopic standard-flash b-wave, the three sib-level means are $65.1 \mu\text{V}$ (II:1), $38.4 \mu\text{V}$ (II:3) and $24.6 \mu\text{V}$ (II:5); the family-level mean is therefore $(65.1 + 38.4 + 24.6) / 3 = 42.7 \mu\text{V}$.

Subject (F6)	Age (y)	Eye	Scot 0.01 b	Scot SF a	Scot SF b	Phot a	Phot b	30-Hz
II:1	29	OD	36.2	26.8	66.4	9.1	43.7	33.8
II:1	29	OS	35.0	26.0	63.8	8.7	42.6	32.7
II:3	26	OD	23.6	17.4	39.1	6.0	28.9	22.4
II:3	26	OS	22.7	16.8	37.7	5.7	27.8	21.5
II:5	24	OD	14.9	11.2	25.3	4.0	18.4	14.2
II:5	24	OS	14.5	10.8	23.9	3.9	17.6	13.6
II:1 mean (both eyes)			35.6	26.4	65.1	8.9	43.2	33.3
II:3 mean (both eyes)			23.2	17.1	38.4	5.9	28.4	22.0
II:5 mean (both eyes)			14.7	11.0	24.6	4.0	18.0	13.9
F6 family mean			24.3	17.8	42.7	6.1	29.5	22.8

Table 5. Spectral-domain OCT measurements (mean of both eyes of the proband). CRT = central retinal thickness (μm) at the foveal centre. EZ width = ellipsoid-zone preservation width across the horizontal foveal scan (μm). ONL = outer-nuclear-layer thickness at fovea (μm). Choroid = subfoveal choroidal thickness (μm) on enhanced-depth imaging. Reference values ($n = 124$ adult controls): CRT 268 ± 21 , ONL 88 ± 10 , EZ width $6,200 \pm 410$, choroid 287 ± 64 .

Family	Gene	CRT (μm)	ONL (μm)	EZ width (μm)	Choroid (μm)	Cystoid oedema	Bone-spicule
F1	USH2A	214	61	3,180	248	Yes	Yes
F2	ABCA4	186	52	4,420	224	No	No
F3	CRB1	321	94	1,940	301	Yes	Yes
F4	RPE65	176	41	620	198	No	Mild
F5	EYS	202	57	2,610	236	No	Yes
F6	CERKL	194	54	3,470	229	No	Yes
F7	cand. PDE6A/NR2E3	221	63	2,840	244	No	Yes
F8	cand. TULP1/FAM161A	209	59	2,470	239	Mild	Yes
F9	DHDDS	217	61	3,020	241	No	Yes
F10	DHDDS	223	64	3,210	246	No	Yes

Table 6. Goldmann visual-field isopter areas (proband, mean of both eyes), in square degrees. V4e = brightest test stimulus; III4e = smaller, dimmer stimulus. % residual = area as a fraction of the age-matched normative V4e area (12,420 deg^2 for adults).

Family	Gene	V4e area (deg^2)	III4e area (deg^2)	V4e % residual	Ring scotoma
F1	USH2A	2,840	510	22.9	Yes (30–55°)
F2	ABCA4	8,140	2,920	65.5	No
F3	CRB1	1,560	210	12.6	Yes (15–40°)
F4	RPE65	410	0	3.3	No (constricted)
F5	EYS	3,920	880	31.6	Yes (20–50°)
F6	CERKL	4,470	1,140	36.0	Yes (25–55°)
F7	cand. PDE6A/NR2E3	2,210	430	17.8	Yes (20–45°)
F8	cand. TULP1/FAM161A	1,910	340	15.4	Yes (15–45°)
F9	DHDDS	2,610	490	21.0	Yes (25–50°)
F10	DHDDS	2,890	560	23.3	Yes (25–55°)

Table 7. Targeted Sanger sequencing of *DHDDS* (NM_024887.4) in families F9 and F10. Seven amplicons covering all coding exons plus ≥ 30 bp intronic flanks (total target 2,833 bp). Genotypes at c.124A>G (chr1:26864711, GRCh37) are shown for each tested individual; segregation is fully consistent with autosomal-recessive inheritance. Phred quality is the mean across bidirectional reads at the variant base.

Family	Individual	Relation	Affected	Genotype c.124	Phred
F9	II:1	proband	Yes	G/G	58
F9	II:2	sib	Yes	G/G	56
F9	I:1	father	No	A/G	54
F9	I:2	mother	No	A/G	55
F9	II:3	sib	No	A/G	53
F10	II:2	proband	Yes	G/G	57
F10	II:4	sib	Yes	G/G	55
F10	I:1	father	No	A/G	52
F10	I:2	mother	No	A/G	54

Family	Individual	Relation	Affected	Genotype c.124	Phred
F10	II:1	sib	No	A/A	50

4. Discussion

Across this 10-family AR-IRD cohort, WES plus targeted *DHDDS* sequencing yielded an unambiguous bi-allelic diagnosis in 8 of 10 probands (80.0%; 95% CI 49.0–94.3%). Two families (F7, F8) remain genetically ambiguous because two unrelated, individually plausible candidate variants segregate with disease in each pedigree, and phenotype does not discriminate. This residual ambiguity rate of 20.0% sits at the upper end of published estimates (Sangermano 2018 reports 11.8%; Carss 2017 reports 17.4%) and underscores the limitation of segregation analysis in small AR pedigrees in which informative meioses are scarce. Functional studies (mini-gene splice assays for the two splice-donor candidates in F7 and F8; iPSC-derived retinal organoid electrophysiology) are planned to discriminate.

The two AJ families F9 and F10 both carry the founder allele *DHDDS* c.124A>G in homozygosity, consistent with the published AJ-non-Finnish carrier frequency of 0.84% in gnomAD v2.1.1 (36 / 4,288 alleles). Their phenotype was milder than the *RPE65* and *CRB1* probands and closer to the *EYS* proband, with scotopic standard-flash b-wave amplitudes of 21.3 μ V (F9) and 23.6 μ V (F10), compared with 11.4 μ V in *RPE65*-associated F4. The *CERKL* family F6 — to our knowledge the first reported South-Asian *CERKL* pedigree — showed an intermediate, well-clustered phenotype (per-sib means 24.6, 38.4 and 65.1 μ V; family mean 42.7 μ V on the scotopic standard-flash b-wave) and residual EZ width of 3,470 μ m. Of note, the youngest sib (II:5, age 24) is already at amplitudes comparable to the *USH2A* proband F1, while the eldest (II:1, age 29) still retains $\approx 1/6$ of the lower-bound normal amplitude. This intra-familial variability is consistent with the documented age-dependent decline in *CERKL*-associated phenotypes.

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Supplementary Materials

The following supplementary tables expand the data summarised in the main paper. They are reported in full for transparency. They do not supersede the per-family summary values reported in Tables 4–7 of the main paper.

Table S1. PCR primers used for targeted Sanger sequencing of *DHDDS*. Amplicon sizes total 2,833 bp.

Exon	Forward primer (5'→3')	Reverse primer (5'→3')	Amplicon (bp)	Ta (°C)
1	GCCTAAGCGTAATCTAGGCT	TTGCAATGTGGTCAGCAAGT	312	60.2
2	ACGTAATCGGTTACGCTTGA	CGTGAACCTCACGCTGAATC	408	59.8
3	TAGCCATGGTACAAGTGCAA	GACATGCGTACGGAATGTCA	487	61.0
4	AAGGCTCATGGAATGCCAAT	CCTTGAAGCGTCATAGGCTT	521	60.5
5	CACGGTACAGTTGAATGGCT	TGTGCAATCGTTAGCCATGA	366	60.0
6	ATTGCCATGGTTCAAGCGTA	CGTAACGGTTAGCATGCCTT	298	59.5
7	GGATCGTAACGGTTAGCATG	TCAAGCGTAACGGTTACGTA	441	60.7
Total	—	—	2,833	—

Table S2. gnomAD v2.1.1 allele counts for each causal variant.

Variant	Total AC	Total AN	Total AF	NFE AF	SAS AF	AFR AF	EAS AF	AJ AF
USH2A c.2299delG	42	250,818	1.7e-4	3.1e-4	9.4e-5	0	0	2.1e-4
ABCA4 c.5882G>A	85	250,082	3.4e-4	5.8e-4	1.2e-4	6.7e-5	0	2.4e-4
CRB1 c.2843G>A	52	246,514	2.1e-4	2.7e-4	4.9e-4	0	0	1.6e-4
CRB1 c.498_506del	0	247,206	NA	NA	NA	NA	NA	NA
RPE65 c.1102T>C	20	250,610	8.0e-5	1.2e-4	0	2.7e-5	0	9.4e-5
EYS c.4350_4356del	0	248,002	NA	NA	NA	NA	NA	NA
CERKL c.769C>T	11	263,084	4.2e-5	6.6e-5	2.0e-5	0	0	4.7e-5
PDE6A c.998+1G>A	0	250,940	NA	NA	NA	NA	NA	NA
NR2E3 c.166G>A	164	247,664	6.6e-4	8.8e-4	3.1e-4	2.4e-4	0	5.1e-4
TULP1 c.1495+1G>A	0	248,810	NA	NA	NA	NA	NA	NA
FAM161A c.1567C>T	28	250,304	1.1e-4	1.4e-4	6.8e-5	2.7e-5	0	9.9e-5
DHDDS c.124A>G	239	248,720	9.6e-4	2.2e-4	0	0	0	8.3e-3

Table S3. Longitudinal full-field ERG (scotopic standard-flash b-wave amplitudes, μV) at baseline and at 12, 24 and 36 months of follow-up for the proband of each family. NB: these longitudinal values are **different** from the per-family means in Table 4 of the main paper, which are cross-sectional and averaged across all affected sibs.

Family	Gene	Baseline	12 mo	24 mo	36 mo	Δ/yr (μV)
F1	USH2A	41.2	36.9	31.8	27.4	-4.6
F2	ABCA4	94.7	88.1	81.0	73.5	-7.1
F3	CRB1	18.4	16.1	13.7	11.2	-2.4
F4	RPE65	12.6	10.8	8.9	7.0	-1.9
F5	EYS	48.3	43.7	38.6	33.4	-5.0
F6	CERKL	66.4	60.1	53.2	46.7	-6.6
F7	cand. PDE6A/NR2E3	31.8	28.4	24.6	20.7	-3.7
F8	cand. TULP1/FAM161A	26.1	23.2	19.9	16.4	-3.2
F9	DHDDS	29.4	26.3	22.8	19.1	-3.4

Family	Gene	Baseline	12 mo	24 mo	36 mo	Δ/yr (μV)
F10	DHDDS	32.7	29.1	25.0	20.7	-4.0

Table S4. For completeness, per-family *median* (rather than mean) ERG amplitudes across affected individuals. The primary analysis reported in the main paper uses the mean (Table 4); medians are presented here as a sensitivity analysis only.

Family	Scot 0.01 b (med.)	Scot SF a (med.)	Scot SF b (med.)	Phot a (med.)	Phot b (med.)	30-Hz (med.)
F1	18.2	11.1	29.2	4.6	21.3	16.0
F2	61.4	41.0	78.1	12.2	60.7	46.6
F3	9.1	6.7	16.0	3.0	14.5	11.7
F4	5.4	4.2	11.4	1.8	9.6	7.3
F5	21.2	14.6	36.8	5.5	25.8	19.1
F6	23.2	17.1	38.4	5.9	28.4	22.0
F7	15.0	9.4	23.8	5.2	23.1	18.3
F8	12.6	8.3	19.4	4.3	18.8	14.8
F9	11.5	7.8	21.0	4.8	20.5	16.1
F10	13.2	8.5	23.2	5.0	21.7	16.8

Supplementary Note S5: extended methods

Sequencing libraries were quantified by qPCR (KAPA Library Quantification Kit, 5 standards 10 pM to 10 nM) with mean library concentration 18.4 nM (range 12.7–24.6 nM). Final pooled flow-cell loading concentration was 1.8 pM with 1.0% PhiX spike-in. Reads were trimmed with Trimmomatic v0.39 (LEADING:3 TRAILING:3 SLIDINGWINDOW:4:15 MINLEN:50). Mean read length post-trimming was 147.3 bp. Duplicates flagged by Picard MarkDuplicates averaged 8.4% (range 6.1–11.7%). BQSR recalibration tables were generated from dbSNP build 138 (chr1–22, X, Y). HaplotypeCaller was run with -ERC GVCF, --standard-min-confidence-threshold-for-calling 30. Joint genotyping with GenotypeGVCFs was followed by VQSR (SNP truth: HapMap 3.3, Omni 2.5, 1000G phase 1; INDEL truth: Mills+1000G gold standard); SNP tranche 99.7%, INDEL tranche 99.0%. Per-sample concordance with high-confidence GIAB calls (NA12878 spike-in lane) was 99.62%.

Variant interpretation: ACMG-AMP 2015 criteria were applied, supplemented by ClinGen Hearing Loss VCEP guidance for *USH2A* and ClinGen Retina VCEP guidance for *ABCA4*, *CRB1*, *RPE65*. Cut-offs: PVS1 for null variants (frameshift, nonsense, $\pm$ 1–2 splice); PM2_supp for absence from gnomAD; PP3 if CADD $\geq$ 20 AND REVEL $\geq$ 0.50 for missense; PS3_moderate if peer-reviewed functional study available; BS1 if $AF_{\text{gnomAD}} > 5 \times 10^{-3}$.

ERG normative database: n = 124 healthy adult controls (62 male, 62 female), age range 18–61 years (mean 36.8 ± 12.1). Recordings were performed in the same laboratory on the same Diagnosys Espion E3 instrument between 2010 and 2016. Test–retest reliability over 4 weeks in 24 controls: scotopic SF b-wave intra-class correlation 0.91. All affected-individual amplitudes reported in this paper are mean of the OD and OS measurements of that individual; family-level values are the unweighted mean across all affected individuals contributing data, i.e. (Σ per-sib means) / (# affected sibs).